

Creating a z/OS Data Set

Creating a z/OS Data Set Using ISPF





DATA SET ALLOCATION AND CATALOGUING

You've already learned that z/OS calls a "file" a *data set* and that it keeps track of the location of data sets using a *catalog*.

The MVS "room" of the house doesn't use filesystems, inodes, or any other familiar way of managing files. IBM developed their own approach to persisting data before UNIX was invented.

Creating a data set (or file) is called *allocating* the data set.

Adding information about the data set to the catalog is called *cataloguing* the data set.

We can perform these operations interactively through ISPF (or plain TSO) or by writing JCL to run batch jobs that perform them.

With Ansible, we use the `zos_data_set`, `zos_tso_command`, and `zos_job_submit` modules from the `ibm_zos_core` collection to invoke these z/OS operations.

Our next step is to create a data set through ISPF.



USING ISPF TO CREATE A DATA SET

```
Menu Utilities Compilers Options Status Help
-----
ISPSP Primary Option Menu
Option ==> |
0 Settings      Terminal and user parameters
1 View          Display source data or listings
2 Edit          Create or change source data
3 Utilities      Perform utility functions
4 Foreground    Interactive language processing
5 Batch         Submit job for language processing
6 Command       Enter TSO commands
7 Dialog Test   Perform dialog testing
9 IBM Products  IBM program development products
10 SCLM         SW Configuration Library Manager
11 Workplace    ISPF Object/Action Workplace
12 z/OS System  z/OS system programmer applications
13 z/OS User    z/OS user applications

Enter X to Terminate using log/list defaults
```

The ISPF panel for allocating data sets is located under the **Utilities** Option.

We navigate to the Utilities panel by entering its option number (in this case, 3) and pressing Enter.

USING ISPF TO CREATE A DATA SET

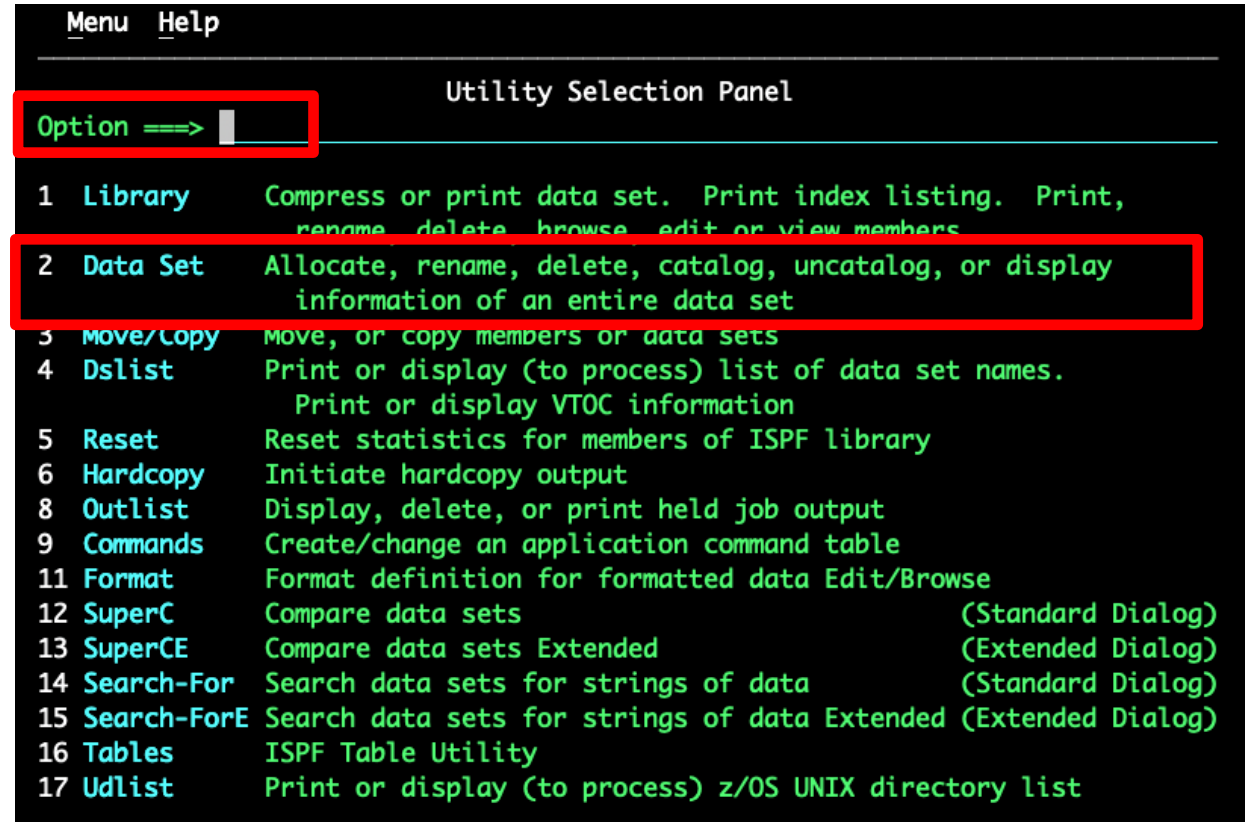
Option 3 from the Primary Option Menu takes us to the Utility Selection Panel.

There are options to manage libraries (PDS and PDSE data sets), basic z/OS data set types (QSAM, etc.), and VSAM data sets.

There are also utilities to compare data sets (SuperC and SuperCE) and to search for data sets containing particular values (Search-For and Search-ForE).

The Tables option pertains to customizing the ISPF Editor. It's out of scope for this class. The Udlist option pertains to Unix System Services (USS), also out of scope here.

You can probably surmise from the name of the option and its description that we're looking for **Option 2 – Data Set utilities.**



USING ISPF TO CREATE A DATA SET

```
Menu Utilities Compilers Options Status Help
ISPF Primary Option Menu
Option ==> █
0 Settings      Terminal and user parameters
1 View          Display source data or listings
2 Edit          Create or change source data
3 Utilities      Perform utility functions
4 Foreground    Interactive language processing
5 Batch         Submit job for language processing
6 Command       Enter TSO commands
7 Dialog Test   Perform dialog testing
9 IBM Products  IBM program development products
10 SCLM         SW Configuration Library Manager
11 Workplace    ISPF Object/Action Workplace
12 z/OS System  z/OS system programmer applications
13 z/OS User    z/OS user applications

Enter X to Terminate using log/list defaults
```

Pressing PF3 or PF12 takes us back where we came from – in this case, the Primary Option Menu.

We can navigate to nested ISPF panels by entering the Option codes for all the nested levels separated by periods. So, we can reach the Data Set Utilities panel directly from the Primary Option Menu by typing 3.2 and pressing Enter.

The 3 navigates to the Utility Selection Panel and the 2 navigates from there to the Data Set Utilities panel.

USING ISPF TO CREATE A DATA SET

Here's the upper portion of the Data Set Utility panel. Functions available here include:

A – Allocate (create) data set

C – Catalog data set

U – Uncatalog data set

D – Delete (uncatalog and unallocate)

blank, S – View detailed or summary information about a data set

V – Navigate to the VSAM Utilities panel

When working with z/OS applications, we use these functions frequently.

```

Menu  RefList  Utilities  Help
-----
Option ==> █ Data Set Utility

      A Allocate new data set
      R Rename entire data set
      D Delete entire data set
blank Data set information
      C Catalog data set
      U Uncatalog data set
      S Short data set information
      V VSAM Utilities

ISPF Library:
Project . . . _____
Group  . . . _____
Type   . . . . _____

Enter "/" to select option
/ Confirm Data Set Delete

Other Partitioned, Sequential or VSAM Data Set:
Name . . . . . _____
Volume Serial . . . _____ (If not cataloged, required for option "C")

Data Set Password . . . (If password protected)
  
```


USING ISPF TO CREATE A DATA SET

```

Menu  RefList  Utilities  Help
                                     Data Set Utility
Option ==> a

  A Allocate new data set          C Catalog data set
  R Rename entire data set        U Uncatalog data set
  D Delete entire data set        S Short data set information
blank Data set information        V VSAM Utilities

ISPF Library:
Project . . . _____ Enter "/" to select option
Group . . . _____ / Confirm Data Set Delete
Type . . . . . _____

Other Partitioned, Sequential or VSAM Data Set:
Name . . . . . qsam1
Volume Serial . . . _____ (If not cataloged, required for option "C")

Data Set Password . . . _____ (If password protected)

```

```

User ID . : IBMUSER
Time. . . : 11:19
Terminal. : 3278
Screen. . : 1
Language. : ENGLISH
Appl ID . : ISR
TSO logon : DB2PROC
TSO prefix: IBMUSER
System ID : VS01
MVS acct. : ACCT001
Release . : ISPF 8.1

```

To identify the data set we want to create, we entered **qsam1** without enclosing the value in apostrophes (single quotes). TSO will change the text to uppercase and prepend the **TSO Prefix** as the HLQ, resulting in data set name **IBMUSER.QSAM1**. We could enter the full data set name within apostrophes: ' **IBMUSER.QSAM1** '.

If we were authorized to create data sets under other HLQs, we could enter the full data set name enclosed within apostrophes, like ' **PROD45.SALES.COMM200** '.

USING ISPF TO CREATE A DATA SET

```

Menu  RefList  Utilities  Help
-----
Data Set Utility

Option ==> a

  A Allocate new data set      C Catalog data set
  R Rename entire data set    U Uncatalog data set
  D Delete entire data set    S Short data set information
blank Data set information    V VSAM Utilities

ISPF Library:
Project . . . _____
Group  . . . _____
Type   . . . . . _____

Enter "/" to select option
/ Confirm Data Set Delete

Other Partitioned, Sequential or VSAM Data Set:
Name . . . . . lab.textfile
Volume Serial . . . _____ (If not cataloged, required for option "C")

Data Set Password . . . (If password protected)

```

Project → High-Level Qualifier

Group → Mid-Level Qualifier

Type → Low-Level Qualifier (Name)

A final comment on data set names and ISPF panel fields before we go on. Notice the section of the panel in the middle labeled **ISPF Library**. The **Project** field refers to the High-Level Qualifier. The **Group** field refers to a Mid-Level Qualifier. The most common convention in z/OS installations is to have one Mid-Level qualifier, although it is valid to have more than one. Providing a single field for Mid-Level Qualifier reflects this convention. The **Type** field is the file name. ISPF panels use this terminology consistently. We could enter those values here instead of in the **Other...Data Set** field. ISPF will convert data set name values to uppercase.

USING ISPF TO CREATE A DATA SET

```
Menu  RefList  Utilities  Help
                                     Data Set Utility
Option ==> a

  A Allocate new data set      C Catalog data set
  R Rename entire data set    U Uncatalog data set
  D Delete entire data set    S Short data set information
blank Data set information    V VSAM Utilities

ISPF Library:
Project . . . _____
Group . . . _____
Type . . . . . _____

Enter "/" to select option
/ Confirm Data Set Delete

Other Partitioned Sequential or VSAM Data Set:
Name . . . . . qsam1
Volume Serial . . . _____ (If not cataloged, required for option "C")

Data Set Password . . . _____ (If password protected)
```

We enter the option we want – in this case **A** for **Allocate new data set**, and press Enter.

USING ISPF TO CREATE A DATA SET

```

Menu  RefList  Utilities  Help

Allocate New Data Set

Command ==>

Data Set Name . . . : IBMUSER.QSAM1

Management class . . . (Blank for default management)
Storage class . . . SCBASE (Blank for default storage)
Volume serial . . . USRVS1 (Blank for system default)
  Multiple Volumes . . . (Enter '/' to select option)
Device type . . . (Generic unit or device address)
Data class . . . (Blank for default data class)
Space units . . . TRACK (BLKS, TRKS, CYLS, KB, MB, or RECORDS)

Average record unit . . . (M, K, or U)
Primary quantity . . . 2 (In above units)
Secondary quantity . . . 3 (In above units)
Directory blocks . . . 0 (Zero for sequential data)
Record format . . . FBA
Record length . . . 80
Block size . . . 27920
Data set name type . . . (LIBRARY, PDS, LARGE, BASIC, *
                           EXTREQ, EXTPREF or blank)

Data set version . . . -
Num of generations . . .
Extended Attributes . . . (NO, OPT or blank)
Expiration date . . . (YY/MM/DD, YYYY/MM/DD
                      YY.DDD, YYYY.DDD in Julian form
                      DDDD for retention period in days)

F1=Help    F2=Split    F3=Exit    F7=Backward  F8=Forward  F9=Swap
F10=Actions F12=Cancel

```

That takes us to the Allocate New Data Set panel, where we can enter the attributes of the data set we want to create.

The data set name is pre-filled from the previous panel.

Other fields are pre-filled with values from the last time the panel was used. We want to override these values for our new data set.

It's a rule of thumb to take installation defaults for Management class, Storage class, and Data class unless you have a specific reason to use different values.

USING ISPF TO CREATE A DATA SET

```

Menu  RefList  Utilities  Help

Allocate New Data Set

Command ==>

Data Set Name . . . : IBMUSER.QSAM1

Management class . . . (Blank for default management class)
Storage class . . . SCBASE (Blank for default storage class)
Volume serial . . . USRVS1 (Blank for system default volume) **
Multiple Volumes - (Enter '/' to select option)
Device type . . . (Generic unit or device address) **
Data class . . . (Blank for default data class)
Space units . . . TRACK (BLKS, TRKS, CYLS, KB, MB, BYTES or RECORDS)
Average record unit (M, K, or U)
Primary quantity . . 1 (In above units)
Secondary quantity . 1 (In above units)
Directory blocks . . 0 (Zero for sequential data set) *
Record format . . . FB
Record length . . . 80
Block size . . . 32000
Data set name type (LIBRARY, PDS, LARGE, BASIC, * EXTREQ, EXTPREF or blank)

Data set version . : -
Num of generations : 
Extended Attributes (NO, OPT or blank)
Expiration date . . (YY/MM/DD, YYYY/MM/DD YY.DDD, YYYY.DDD in Julian form DDDD for retention period in days)

F1=Help F2=Split F3=Exit F7=Backward F8=Forward F9=Swap
F10=Actions F12=Cancel

```

DASD space is allocated in *extents*. The primary extent is reserved for the data set when it is allocated. As the data set grows, secondary extents are created to accommodate the additional data.

This example specifies space in terms of tracks, and allocates one track for the primary extent. Secondary extents will consist of 1 track each.

You can also specify the size in terms of cylinders, blocks, records, bytes, kilobytes, or megabytes. The brain in a jar sees everything in terms of IBM 3390 DASD devices, which have tracks and cylinders.

There is an installation-defined limit to how many extents can be added. When an application attempts to exceed this limit, it fails with an abend code of 0B37, 0D37 or 0E37 (these abend codes are equivalent).

USING ISPF TO CREATE A DATA SET

```

Menu  RefList  Utilities  Help

Allocate New Data Set

Command ==> _____ More: +

Data Set Name . . . : IBMUSER.QSAM1

Management class . . . (Blank for default management class)
Storage class . . . . SCBASE (Blank for default storage class)
Volume serial . . . . USRVS1 (Blank for system default volume) **
Multiple Volumes . . . (Enter '/' to select option)
Device type . . . . . (Generic unit or device address) **
Data class . . . . . (Blank for default data class)
Space units . . . . . TRACK (BLKS, TRKS, CYLS, KB, MB, BYTES
                             or RECORDS)
Average record unit . . (M, K, or U)
Primary quantity . . . 1 (In above units)
Secondary quantity . . 1 (In above units)
Directory blocks . . . 0 (Zero for sequential data set)
Record format . . . . FB
Record length . . . . 80
Block size . . . . . 32000
Data set name type . . . (LIBRARY, PDS, LARGE, BASIC, *
                           EXTREQ, EXTPREF or blank)

Data set version . . : -
Num of generations : -
Extended Attributes . (NO, OPT or blank)
Expiration date . . . (YY/MM/DD, YYYY/MM/DD
                       YY.DDD, YYYY.DDD in Julian form
                       DDDD for retention period in days)

F1=Help    F2=Split    F3=Exit    F7=Backward  F8=Forward  F9=Swap
F10=Actions F12=Cancel

```

Directory blocks pertain to libraries – that is, Partitioned Data Sets (PDS, PDSE types). For a "regular" sequential data set, specify zero directory blocks and leave the **Data set name type** field blank.

USING ISPF TO CREATE A DATA SET

```

Menu  RefList  Utilities  Help

Allocate New Data Set

Command ==> _____ More: +

Data Set Name . . . : IBMUSER.QSAM1

Management class . . . (Blank for default management class)
Storage class . . . SCBASE (Blank for default storage class)
Volume serial . . . USRV51 (Blank for system default volume)
Multiple Volumes . . . (Enter '/' to select option)
Device type . . . (Generic unit or device address)
Data class . . . (Blank for default data class)
Space units . . . TRACK (BLKS, TRKS, CYLS, KB, MB, or RECORDS)

Average record unit . . . (M, K, or U)
Primary quantity . . 1 (In above units)
Secondary quantity . 1 (In above units)
Directory blocks . . 0 (Zero for sequential data set)

Record format . . . FB
Record length . . . 80
Block size . . . 32000

Data set name type . . (LIBRARY, PDS, LARGE, BASIC, EXTREQ, EXTPREF or blank)

Data set version . . : -
Num of generations . : -
Extended Attributes . . (NO, OPT or blank)
Expiration date . . . (YY/MM/DD, YYYY/MM/DD
                      YY.DDD, YYYY.DDD in Julian form
                      DDDD for retention period in days)

F1=Help    F2=Split    F3=Exit    F7=Backward  F8=Forward  F9=Swap
F10=Actions F12=Cancel

```

We're creating a data set with fixed-length records 80 bytes long, grouped into blocks, hence "record format" is "FB".

QSAM reads/writes **blocks** of data, also called *physical records*, and presents the data to application programs one *logical record* at a time. We specify 80 bytes as the logical record length and a block size that is an even multiple of the logical record length.

All this is virtualized today, but when DASD were physical devices the difference in I/O delay could be significant based on these settings.

The remaining fields pertain to other types of data sets or to additional attributes that may be associated with a data set.

USING ISPF TO CREATE A DATA SET

```

Menu  RefList  Utilities  Help
                                     Data Set Utility
Option ==> █ Data set allocated

A Allocate new data set      C Catalog data set
R Rename entire data set    U Uncatalog data set
D Delete entire data set    S Short data set information
blank Data set information  V VSAM Utilities

ISPF Library:
Project . . . _____ Enter "/" to select option
Group . . . _____ / Confirm Data Set Delete
Type . . . . . _____

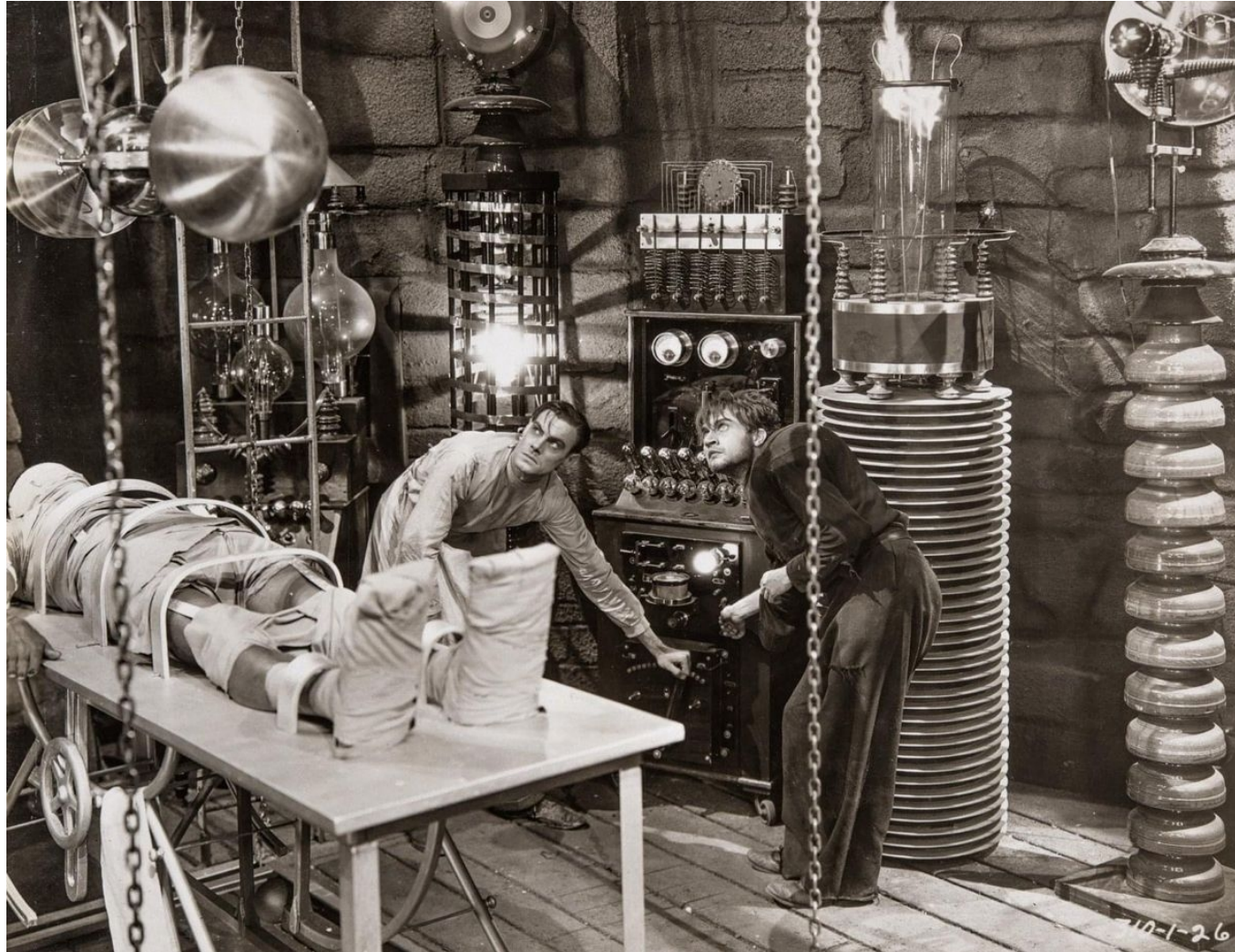
Other Partitioned, Sequential or VSAM Data Set:
Name . . . . . QSAM1
Volume Serial . . . _____ (If not cataloged, )
Data Set Password . . . _____ (If password protect

```

When we press Enter, TSO allocates the data set and returns us to the Data Set Utility panel, where we can work with more data sets if we need to. It displays the result of the allocation request in the upper right-hand corner.

The fields on the ISPF panels correspond to the same attributes we define as parameters on DD statements in JCL or as vars in an Ansible task.

LAB Z/OS 5: CREATE A DATA SET USING ISPF



WHAT YOU KNOW AND WHAT YOU CAN DO

- You can run TSO commands from the READY prompt.
- You can navigate around ISPF panels.
- You can create a QSAM dataset using ISPF.

WHAT'S NEXT

- Job Control Language (JCL)